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ORIGINAL RESEARCH

How Social Should AI Be?

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Abstract
Social AI systems have the potential to outperform humans in certain ways—raising the question of whether we should have AI systems (including robots) replace humans in the labor market (i.e., not only in non-social contexts). This article will first put forward three sets of *pro tanto* reasons why social AI systems should not be implemented, then it will consider three possible situations in which these reasons may be overridden, arguing that the use of social AI is only justified in a very circumscribed set of cases. Moreover, in situations in which replacement is justified, the AI systems' social and informational capabilities should be limited. And finally, where social AI systems are utilized, they should be made to comply with certain social codes.

1 Introduction

Recently, Fröding and Peterson argued that AI systems should be "programmed to behave as-if they were friends of the individual human", as opposed to creating systems that function as slaves or utility maximizers.[1] The idea of making friendly AI systems is not new. Indeed, it has been part of the science-fiction literature and film traditions for decades. Recent technological developments have arguably contributed to a broader academic interest, but the serious academic discussions go back to at least 1999 and topics related to friendly or social AI have been discussed ever since.[2]

[1] Fröding et al., (2021), p. 209.

[2] Breazeal (1999).

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While "friendly AI" is sometimes used to denote the property of being value-aligned, which may have nothing to do with how the system behaves socially, my focus is on the notion of social AI, of which friend-like systems (in the sense used by Fröding and Peterson) are a subset.

Debates on this topic have been especially prevalent in the literature on robotics. For example, researchers have asked whether we can be friends with robots;[3] whether robots can be good colleagues;[4] whether robots should be teachers;[5] whether robots should be allowed to punish us;[6] and whether a robot can be a citizen or acquire other rights associated with personhood.[7]

In this paper, I aim to contribute to this growing debate by focusing on a question that, although it underpins all the above queries, has been ignored—or at least set aside—namely: should we have social AI systems? That is, I want to ask whether we should embark on a radical change in society (the basis for which arguably already started with other systems and devices, such as smartphones and social media). To answer this question, it may *prima facie* seem as if we need a thorough definition of social AI systems. However, for the present purpose, it is sufficient to say that the class of objects that I am interested in are systems (whether purely algorithmic or in an embodied machine) that *can engage socially with humans*, which in turn will require an ability to exchange data or information with those humans. The prototypical example of social AI systems is of course various smart social robots (such as carebots, robotic teachers, or sex robots), but the argument also applies to advanced large language models or systems that have sophisticated natural language processing, such as ChatGPT, Amazon's Alexa, Google Assistant, Apple's Siri, or Replika's services. In the argument, I will also discuss robots that imitate animal-human social relations (such as PARO). Many of these systems vary in their social complexities and while a good definition should qualify the precise threshold between social and non-social systems, in the given case such precision is not necessary for at least three reasons: First, because it will be clear from my argumentation what I broadly have in mind. Second, the borderline cases will be less problematic, so although those cases are conceptually and definitionally relevant, they matter less here for my normative purpose. Third, some of the arguments I will present have different scopes and hence work for different thresholds of what can be called "social AI systems."

Nevertheless, there is one important distinction that I must note: the difference between systems that provide opportunities for socialization between humans (and perhaps contribute to moderating those interactions)—such as social networks—and systems that themselves provide social interaction. Although some of my arguments will hold for social media as well, what I am addressing are systems that interact socially with humans, not systems that facilitate interactions between humans and other humans.

[3] Danaher (2019).

[4] Nyholm and Smids (2020).

[5] Sharkey (2016).

[6] Wagner and Jois (2022).

[7] Rainey (2016); Coeckelbergh (2010). Famously, Saudi Arabia granted citizenship to the robot Sophia, in 2017 (Walsh, 2017).

In the broad debate on AI, many presume that AI systems (including robots) will replace humans in many jobs in the labor market.[8] However, we may have reasons to avoid such a future—and we are still in a position to affect the future through political decision-making and other means.[9] In this paper, I argue that there are strong reasons to limit the replacement of human-to-human social interaction, available through the job market, with human-to-AI social interaction. To establish this, I provide three sets of *pro tanto* reasons to avoid social AI systems in the next section.

However, there are situations in which these concerns are overridden—that is, when social AI systems are justified. I discuss such cases in the third section. I argue that under the conditions when social AI systems are justified, there are still strong limits and constraints on information and data processing. That is, when the usage of social AI systems is justified, their social interaction should be purpose-limited and we should minimize their information and data collection, dissemination, processing, and risks. Indeed, my strongest arguments are those in favor of disconnected AI systems, contrary to the growing trend of cloud-based, Wi-Fi, or otherwise connected units that are part of the broad concept of the Internet-of-Things. Simply put, in those limited cases when social AI systems are justified, their information usage, processing, and dissemination should be limited, which speaks in favor of disconnected AI systems. To be clear, the property of data limitation attaches to data collected in social situations (necessarily, robots that engage with humans may need to have various memory functions, e.g., to ensure safety).

Moreover, there may be situations in which such systems are not replacing humans, but filling a gap, which may change the ethical analysis. I turn to that issue in the fourth section. This provides us with cases when, potentially, social AI systems may be justified. However, *pace* Fröding and Peterson, and others, I argue that the default standard ought to be that AI systems should be professional rather than friendly. Friendly behavior is very contextual. It requires knowledge about the individual. Behaving professionally, on the other hand, is a relatively well-codified (or codifiable) behavior. I discuss this in the fifth section. The article ends with a summation of my findings.

Before turning to the discussion, I need to comment on some limitations. First, I will not cover all ethically relevant aspects of these systems. In particular, I will set aside issues relating to the effects on the labor market. This should in no way be seen as an indication that these issues are not important. They are, however, largely orthogonal to the issues that I wish to discuss (i.e., I am not concerned with issues such as unemployment, but rather I am concerned with the problems associated with social interactions with AI systems, which follow from replacing humans with AI systems in social contexts). Moreover, it is not *prima facie* clear whether the effects on the labor market are good or bad since that depends on various complicated empirical issues (such as whether we can tax work by AI systems properly and whether there can be other meaningful occupations for humans in case the labor market shrinks).

[8] See, e.g., Fröding, et al. (2021). For a prediction based on expert surveys, see, e.g., Grace et al. (2018)—but see Lundgren (manuscript) for a critique. Others, such as Autor (2024), argue that with the right policies, AI can be used to rebuild the middle class.

[9] Sullins et al. (2008).

Second, I will not engage with examples in the distant future. Rather I will stick fairly closely to the current technical capabilities and capabilities that are not too far from current achievements so that we can judge their future plausibility.

2 Pro Tanto Reasons Against Social AI Systems

In this section, I will present three reasons against the use of social AI systems. Some of these reasons include a potentially overlapping cluster of arguments, risks, and so forth. Moreover, while there are further reasons available in the literature, I focus on these three reasons because they are sufficient for my argument (even if, as I noted earlier, they address different scopes of or thresholds for what it means for an AI system to be social). This is not to say that any of the reasons I have omitted are irrelevant, but their adduction is strictly unnecessary.

2.1 Human Social Interactions Are Better

The first worry addresses an argument by Fröding and Peterson.[10] They argue that as human-to-human contact is replaced by human-to-AI interaction, humans will need AI systems to provide training for humans in pro-social behavior. However, as I noted in the Introduction, we are still in a position to choose our future, which in the context of their argument raises the question of why we should train AI systems to engage in social situations when we could engage in social situations with humans directly. Simply put, in this subsection, I will argue that ordinary human-to-human social interactions are better for humans than human-to-AI interactions (i.e., by ordinary interaction I mean to exclude edge cases of bad actors).[11] In what sense is the latter better? I think that there's value in human interaction that AI systems cannot replace. There is a broad debate on this in the robotics literature, but I will side-step that because I want to present a new argument for this position.[12]

As a starting point, let me note that "[a] necessary condition of friendship, according to just about every view […] is that the friends each care about the other, and do so for her sake; in effect, this is to say that the friends must each love the other."[13] Undoubtedly, caring must also be authentic, which is at the core of my argument. Put otherwise, inauthentic caring is not caring. Hence, I will argue that an authentic social relationship is better than a non-authentic social relationship (arguably, for some social relationships—such as friendship—authenticity might even be conceptually necessary).

[10] Fröding, et al., (2021).

[11] One might worry that this weakens the argument since I set aside reasons for replacing humans with social AI systems. However, in this section, I address reasons against replacing humans with social AI systems. In Sect. 3, I will balance these reasons *against replacement* against *reasons in favor of replacement*. I will discuss and evaluate the possible gains of replacing bad human actors with social AI systems, in Sect. 3.2.

[12] See, e.g., Sharkey and Sharkey (2012); Sparrow and Sparrow (2006).

[13] Helm (2021). References were removed from the citations.

To show this I will introduce a version of Nozick's experience machine. In Nozick's thought experiment, we consider a situation in which we have the option of plugging into a machine that give us any experience we desire. While desire-satisfaction is the optimal goal according to some ethical theories, Nozick argues against plugging into the machine, suggesting that part of the problem of the experience machine is that it is not real. That is:

> Perhaps what we desire is to live (an active verb) ourselves, in contact with reality. (And this, machines cannot do for us.)[14]

This analysis showcases the argument that I want to put forward here: AI systems cannot give us *real social interaction* because, like the experience machine, there is nothing behind the patterns of behavior expressed by current AI systems.[15] Simply put, if we value authenticity over non-authentic utility in the experience machine, then we should value authenticity over non-authentic utility when it comes to social AI systems as well (i.e., even if they have other benefits).[16]

We also know from empirical tests that a majority of people have a preference for real life rather than the experience machine.[17] Nozick's argument, quoted above, can be expressed as a desire for authentic experiences. That is, happiness alone or what we might call *non-authentic desire-satisfaction* is insufficient, it also matters that "our state of mind is appropriately related to how things are".[18] A recent empirical study aimed to test whether people would reject the experience machine against the possibility of confounding factors and the data seems to support Nozick's original thesis that people value "living their lives in contact with reality".[19] Of course, the argument here is not purely based on an empirical consideration of preferences, there is also a normative value in authentic relationships (as I will elaborate below).

However, the case of replacing humans with social AI systems is not fully symmetric to the experience machine. Indeed, interaction with a social AI system is real, even if it does not have emotions. However, my point here is that we can apply the argument to social interactions and if we value authentic social interaction in the same sense that we value authenticity of experiences, then we can conclude that human interactions have a benefit viz-á-viz interactions with AI systems. Why? Because we know that AI systems' social interactions are non-authentic in the sense that they are non-reciprocal. To ground this claim, it would be optimal to define "social rela-

[14] Nozick (1974, p. 45).

[15] I concur with, for example, Bender et al. (2021) that large language models are simply stochastic parrots. *Mutatis mutandis* for other current AI systems.

[16] While Nozick's >40,000 citations include a few hundred related to this topic, I have not found anyone making the exact same argument. The closest I found was Sætra (2022) who asks "How is the experience machine connected to the social robots, then? My argument is that such a machine is similar to employing social robots, which produces subjective well-being based on relationships and experiences produced by machines. If we suggest that deception is justified if subjective well-being is enhanced, how could we object to connecting the old with dementia to such machines?" (p. 38).

[17] De Brigard (2010); Hindriks and Douven (2018); and Weijers (2014).

[18] Haybron (2020).

[19] Hindriks and Douven (2018, p. 294).

tionships." However, because social relationships are dynamic and context-varied I will not try to define it here. Instead, we can start by considering a pro-typical social relationship and one of the most important relationship—namely, friendship. Friendship is commonly thought to require mutual caring, intimacy, and shared activities.[20] It is fair to say that these concepts are also relevant for romantic engagements and in good-faith family relations. If we based on this consider what one might call mere "social acquaintances," that is, people whom we might engage in some aspects of shared activities (i.e., socialization), it is fair to say that an AI system would not satisfy even a slim reading of *sharing in activities*. Why? Because their engagement is non-reciprocal—that is, they are merely trained or programmed to do so. (Of course, this is not to say that all human-to-human interactions are authentic. However, arguably, most of them are.)

Of course, we can question whether people, in reality, would *know* that. Indeed, consider the case of Google's AI LaMDA, which a Google engineer claimed was sentient, something that was denied by Google and broadly rejected by the scientific community.[21] Yet, even if there were to be confusion—even amongst people who should know better—that does not affect my argument because, in Nozick's argument, it is presumed that if you are in the experience machine, then you do not know that you are in the experience machine either. Indeed, one may argue that the situation is worse when people mistakenly believe that an AI system is providing a reciprocal relationship. This mistaken belief is problematic for at least two reasons: First, because it may become a factor in moral decision-making (in which we are influenced by factors that should not be considered), and second, because the system can manipulate us more broadly (e.g., granted that we have limited emotional capacity, it would be bad if we were to spend emotional resources on objects that deserve no sympathy).

John Danaher attempts to counteract this type of authenticity-based argument by arguing that:

1. It is possible for the mutuality and authenticity conditions to be satisfied in our friendships with our fellow human beings (assumption).
2. The only grounds we have for thinking that the mutuality and authenticity conditions are satisfied in our friendships with our fellow beings are the performative representations that they make to us (i.e., these are the only epistemic grounds we have for believing in human virtue friendships).
3. These epistemic grounds for believing that the mutuality and authenticity conditions are satisfied in our virtue friendships with our fellow human beings can also be satisfied by robots (they can consistently perform mutuality and authenticity).
4. Therefore, it is (technically) possible for the mutuality and authenticity conditions to be satisfied in our friendships with robots.[22]

[20] See, e.g., Helm (2021) for an overview.

[21] See, e.g., Vallance (2022).

[22] Danaher (2019, pp. 12–13). I have re-numbered Danaher's argument from the original, in which 1 is 4, 2 is 5, etcetera.

As argued above, an individual's knowledge is not a factor in my argument. Moreover, Danaher's argument fails because premises 2 and 3 are jointly false. That is, our epistemic ground for believing that authentic conditions are satisfied is not based purely on performative acts, but also in virtue of the fact that the people performing them are humans. We can draw certain conclusions about other agents' inner lives by virtue of some relevant similarities to those agents viz-á-viz their inner lives. Unlike other humans, we have no reason to think that robots or AI systems are similar to us in *this sense*, nor is there anything else that gives us reason to ground a belief that their behaviour corresponds to genuine authentic, reciprocal, social interaction. (Moreover, if we indeed lose vigilance about the truth of the feelings we have, then that exposes us to abuse, something I will discuss in the next subsection.)[23]

The above argument has some *prima facie* flaws. First, it rejects various forms of skepticism. However, if we accept hardcore skepticism, then Danaher's argument also fails. Second, it may seem as if it depends on a set of common-sense assertions mixed with some anthropocentrism. We might worry that such assertions are weak. However, although seemingly problematic, we do not need those assertions to be strong because what Danaher is saying—and what his argument depends on—is that there are "no grounds". Thus, all that we need the argument to do is to supply some non-zero credence for a reason for thinking that there is a difference-maker beyond pure behavior.

However, this might be *prima facie* complicated by the possibility of philosophical zombies—that is, agents that are just like humans but have no inner life (not flesh-eating monsters).[24] One might think that the possibility of zombies affects the non-zero credence above. However, the fact that some believe in (the possibility of) zombies does not mean that we need to assign any credence to the thesis that there *actually* are zombies (i.e., perhaps we can reasonably exclude the actual existence of zombies, even if we cannot exclude their possible existence). While the zombie case concerns a somewhat different question—that is, that of sentience—we can easily modify the discussion to fit with the relevant case.[25] Even if we recognize a non-zero probability that there actually are human zombies and a non-zero probability that AI systems are genuinely social, a probabilistic version of the argument still would not work. Why? Because the probabilities would be at the reverse endpoints (close to 100% that each human is a non-zombie and close to 0% probability that a current AI system is authentically social). Of course, for my counter-argument to hold, we do not even need such extreme differences. All we need is for there to be a non-zero credence for a reason for thinking that there is a difference-maker beyond pure behavior.

One may question some of the claims that I have made by looking to the distant future. However, keep in mind that my argument is for the near future (technologically defined). Moreover, one may think that social interaction with robots or other AI

[23] Danaher has defended versions of this thesis in several publications. More recently, he addresses critics of a version of the argument that is concerned with moral status (which is of no relevance here), see Danaher (2021).

[24] Chalmers (1996).

[25] See Danaher (2020a) for a discussion on zombies and moral status of robots, which is based on similar reasoning as his case for friendship.

systems could be of value in cases when people are lonely (for example) and that, in the absence of humans, social training with machines would be better than no social training at all. However, here I am talking about reasons against *replacement*. The question of whether AI systems can play a role in filling a gap is thus a different class of question. I will return to the question of whether social AI systems can fill a social gap in Sect. 4.

Although we can discuss how strong this argument is, it provides a *pro tanto* reason against social AI systems—irrespective of how the concept is defined along the lines sketched in the Introduction.

One possible counterargument against this conclusion is that the authenticity of human-to-human interaction is itself just a *pro tanto* reason for thinking that it is better than human-to-AI-system interaction since there are other things we value about social interaction. For example, it is plausible that AI systems can be made to be more reliable than humans (or have other kinds of non-social benefits). However, this is precisely symmetric to the experience machine since the experiences in the machine are *better* than the authentic experiences (setting authenticity aside), but non-authentic desire-satisfaction is insufficient. Thus, although we could construct AI systems that may have benefits viz-á-viz humans, the relation would still be inauthentic, which provides a *pro tanto* reason against them.

Moreover, although it may be the case that AI systems could be constructed to be more reliable—setting aside the complexity of ensuring reliance in differing contexts—that does not necessarily imply that we would rely upon them or that we can or should trust them. (As I will discuss in upcoming subsections, such trust or reliance may itself be problematic.)

Lastly, there are other arguments against replacing human relationships with AI systems. Indeed, some worry about replacement simpliciter, for example, because it might compete with time spent on human relationships.[26] However, such arguments are only valid if that is worse, which is what I have argued for in this subsection.

2.2 Informational Risks

We are already in an informational landscape that collects, processes, and disseminates large quantities of user data and information. This is worrisome both on an individual and a collective level. Information and data collection is most obviously worrisome because it may be detrimental to our (condition of) privacy or infringe or violate our right to privacy. While some think that the latter worry can be resolved through informed consent, there are clear limits to the ethical modulation of consent practices.[27] Moreover, even if we think that consent makes information and data gathering ethically permissible viz-a-viz the right to privacy, such practices can still be detrimental to an individual's (condition of) privacy. Furthermore, there are also suggestions in the literature that substantial informational risks themselves can con-

[26] See, e.g., Bryson (2009). It is worthwhile to note that Bryson draws on Dunbar's number—a specific limit for how many social relationships a human can have—a thesis that recently has been disproven (Lindenfors et al., 2021).

[27] See, e.g., Nissenbaum (2011); Solove (2013); and Lundgren (2025).

stitute a violation of, or infringe upon, someone's right to privacy (even if our privacy is not affected).[28] Lastly, the worries are not limited to privacy concerns. We might, more directly, worry about informational harms, or that information and data are distributed in ways that are unjust or lead to injustice, inequality, biases, or negative influences on our autonomy.[29] In particular, we might worry about how big data and information aggregation technologies are used to influence democratic elections.[30]

Arguably, society is already suffering from too much information and data collection and the different forms of problematic usage that follow from it. This is a direct argument against social AI since social AI systems would continue and expand an already problematic process. However, it is important to note that social AI systems would constitute more than a continuation of the already existing practices; they would imply an *increase* in our informational risks and a domain change. The occurrence of AI systems that increase the information collected within our homes is a fairly recent development. Beyond smartphones, systems such as Alexa (a smart voice-controlled assistant) and various sensor systems added to ordinary devices (TVs, kitchen appliances, clocks, etc.) are recent additions that provide surveillance (broadly understood) within our private homes, rather than being limited to our online lives or various public venues.

It is adding to an already problematic trend, and the scope of surveillance is at risk of expansion if social AI systems are used in a context where we are still somewhat free from constant surveillance. Social AI systems may turn our homes, classrooms, and workplaces into complete surveillance systems, in which there is no room for anonymity. This is particularly worrisome for robotic systems, which add the potential for more sensor inputs so that it becomes less transparent when we are engaging with—and, therefore transferring information to—an AI system. These worries are well-known in the robotics literature.[31]

Again, regardless of how the concept of *social AI* is defined, informational and data risks provide a *pro tanto* reason against the usage of social AI systems. Moreover, beyond the fact that social AI systems' inherent informational risks provide a reason against using them, it is also clear that in cases when replacing humans with AI systems is justified, the systems should be closed, self-contained systems, rather than open, interconnected systems, to minimize the risks following from further dissemination.[32]

Lastly, two types of counterarguments deserve to be mentioned. First, we can note that humans also pose informational risks, so it is not clear that replacing humans with AI systems increases informational risks. Second, and relatedly, we can imagine

[28] See, e.g., Lundgren (2021a, b). This idea is grounded in intuitions that underpin analyses of the right to privacy in terms of control—for example, the idea that camera surveillance is a violation of one's privacy right even if no one accesses the content. What Lundgren shows is that if we take Hansson's (2013) *pro tanto* right against risks exposition to extend to the right to privacy, then we can explain the underlying intuitions even if we accepted a limited-access conception of the right to privacy.

[29] See, e.g., van den Hoven (1997); Lundgren (2020); and Véliz (2020).

[30] Lundgren (2020); Véliz (2020). Although this happened on social media, it illustrates a risk that is also entailed in social AI systems.

[31] See, e.g., Calo (2012); Sharkey (2016).

[32] Cf. Glancy (2012).

AI systems that are built to be super secure. Many might prefer to share sensitive information with secure systems, rather than humans.

I doubt the possibility of creating a super secure system that is used broadly by ordinary individuals. There is a reason why top-secret documents sometimes require handling that only allows access in secure rooms. This kind of security is simply not possible if AI systems are supposed to be used in ordinary settings with ordinary people.

More importantly, the idea of secure information systems also depends on manufacturers respecting the end user's privacy and adhering to strict data protection policies. That is a pipe dream. It is simply not consistent with how the information market is currently functioning and, more importantly, recent information security leaks have shown that despite our best attempts, information is arguably always at risk.[33] Moreover, although humans gossip and can spread information using information systems, the current AI systems are, as per their definition, digital, which means that their information is ready and available for dissemination. Hence, I believe there is good reason to worry more about informational risks from AI systems than from humans.

2.3 Malevolent Agents and the Risk of Manipulation, Security, and Safety

In this section, I will address safety and security concerns. The basic worry here is that devices can be abused and that the presence of social AI would result in a situation in which individuals lower their bar of suspicion towards the potential of abusive systems.

Malevolent agents can create such risks in two ways: first, they can create systems that are designed to be abusive; and, second, they can make use of security flaws in a "non-abusive system" to make it abusive. Given that AI systems are already capable of manipulating human behavior,[34] and that computer systems are already better than close acquaintances at determining an individual's personality,[35] if "friendly" social AI systems were to become common, then there is an obvious risk that humans would start trusting social AI systems, which makes us easy targets for manipulation. That is, there is a risk of naïve trust, which can easily be abused by malevolent agents by either hacking systems previously known to be "friendly" or creating systems that are seemingly friendly but serve malevolent purposes. In the robotics literature, worries have been raised about how robots can use human behavioral patterns (such as crying and laughing) to trick people into thinking there is a reciprocal relation.[36] In response to this, Danaher argues—based on an argument like the one I refuted in Sect. 2.1—that these cases are not problematic (and, in fact, not deceptive at all) if the systems are well-intentioned.[37] Even if we accept Danaher's conclusion that these forms of

[33] For information markets, see, e.g., Véliz (2020), and for information security risks, see, e.g., Lundgren and Möller (2019).

[34] See, e.g., Dezfouli et al. (2020).

[35] Youyou et al. (2015).

[36] Turkle (2007).

[37] Danaher (2020b).

deception need not be directly harmful in themselves, my argument here points to a more indirect long-term worry: Even well-intentioned systems could be problematic because they could lead to a situation when we would be more easily misled by actors with bad intentions.[38] That is, well-intentioned systems can make us more sensitive to abuse from non-well-intentioned systems.

It is reasonable to worry about how smart systems can abuse our political and economic freedoms, as algorithms already have.[39] These types of problems are also complicated by the fact that the distinction between acceptable and unacceptable norms about how to handle data and information (i.e., collection, creation, distribution, and processing) are muddled by the praxis of many of the leading IT companies. That is, while the hacking of a social AI system with the intent to gain access to data and information is clearly unethical, we might worry that the norms around information and data distribution on the information market—such as data monetization—in combination with too lax legal protections and/or enforcement creates an opportunity for further abuse and greater information security risks. For example, the role of big data (collection and processing) also increases the risks of information leakage and abuse.

Beyond concerns related to informational risks, social AI systems (whether hacked or due to explicitly malicious design) can also be used to physically harm individuals.[40] We can imagine a social robot being used to commit violent crimes such as abduction or murder. Just as ransomware is used to force individuals, companies, and organizations to pay hefty sums of money, social AI could be used for similar purposes when much more is at stake. For example, social robots could be designed (or hacked) to use (the threat of) violence to force the individual to pay a ransom or perform some action.

Lastly, hacking also illustrates how safety concerns go beyond system failures, which is a relevant concern, especially for systems intended to deal with complex tasks involving humans. While safety checks and tests can potentially diminish the risks of accidents, it is difficult to protect against all possible forms of hardware and software manipulation.[41]

In summation, this subsection, just like the previous section, supplied *pro tanto* reasons both against human-AI replacement and in favor of a system with limited social functions (when such systems are justified).

[38] In ibid., Danaher recognizes that certain forms of deception are genuinely problematic. For example, he agrees with Kaminsky et al. (2017), that a robot that signals satisfaction of sensory privacy by avoiding eye contact, should not have a separate camera focused on you.

[39] See, e.g., Véliz (2020).

[40] Hansson et al. (2021).

[41] Here one might note how the famous Volkswagen emissions scandal, in which engineers designed Volkswagen's vehicles to pass emission tests, was finally uncovered by the EPA in conjunction with California's Air Resources Board (see: https://www.epa.gov/vw/learn-about-volkswagen-violations).

3 Are the Reasons Against Social AI Overridable?

In this section, I will discuss cases that potentially override the concerns that I discussed in the previous section. What we need to look for is situations when we may have reason to replace humans with AI systems, and in such cases, I will ask whether there are reasons to override any of the *pro tanto* reasons against social AI defended in the previous section.

There are three relevant types of contexts in (or tasks for) which we may have reason to replace humans with AI systems. The first is tasks that are harmful to humans. Indeed, if a task is harmful, then we have a strong reason to replace humans with AI systems. Second, there are tasks for which AI systems outperform humans. If task performance is sufficiently important, then that may supply a reason for replacing humans with AI systems. Third, there may be situations in which AI systems are more cost-efficient than humans. *N.B.*, while these contexts/tasks are not mutually exclusive, it is important to note that an AI system can be more cost-efficient without being better at a given task (e.g., because the cost of running the system is minimal once the investment costs are covered).

In this section, I will discuss these three categories in three subsections. In all cases, one may wonder under which conditions the reasons in Sect. 2 are overridable, but as you will see, I can, for most AI systems, avoid that issue.

3.1 Harmful Tasks

Harmful tasks exemplify a potentially strong reason to replace humans with AI systems, such as robots—and an example that also provides a strong reason against some of the worries of the previous section because when harm and especially when the risk to someone's life is at stake, we tend to think that many other worries are less substantial.

However, the standard example of automated systems replacing harmful tasks is the classical industrial robots, which do not raise the type of worries that I engaged with in the previous section. Indeed, industrial robots are extremely limited rule-following machines. That is, traditional forms of harmful tasks can be performed by systems with very limited informational capabilities (i.e., the type of machines that we normally would not qualify as social).

However, there are some more complicated examples. First, robots used, for example, in military settings clearly need communication functions for everyone's safety and quick responsiveness. Nevertheless, that does not mean such functionality cannot be limited. Moreover, they should clearly not be social in a more extensive sense (e.g., inquiring about our social lives). Second, a more challenging case is that of sex robots. On the one hand, if sex robots were to replace human sex workers and thereby minimize the widespread abuse and suffering in that business, that could create tremendous benefits. On the other hand, some worry about sex robots affecting the end-users adapting to a more objectified behavior, with negative consequences for human-to-human relationships;[42] and if human-to-human relationships are decay-

[42] Bisconti (2021).

ing, it is not unreasonable to think that this would lead to an increase in demand for human sex workers.[43]

This is, of course, a highly complicated issue that depends partly on further empirical studies and many more ethical considerations than the issues I have listed above,[44] which are focused only on replacement. However, what we can say is that even if sex robots are justified, they should be closed information systems with limited information gathering. Surprisingly, a quick search reveals that some of these systems are cloud-based, which is highly questionable.[45]

Of course, one may think that "sex buyers" should have a limited right to privacy (in the context of buying sex) because the right to privacy should not protect against illegal or abusive actions.[46] While this is a complicated question, it is important to recognize that a sex robot, unlike a human, cannot be abused. Sensible worries about the usage of sex robots must be phrased in terms of the effects it has on the user and the broader society (e.g., that it leads to abuses of humans, or negatively affects sexual and romantic relationships through norm change). None of this seems to justify overriding someone's right to privacy. Indeed, if sex robots are justified, they are justified in virtue of contributing to an improvement to the current "sex business," which means that they lead to less rather than more abuse.

In conclusion, cases in which AI replacement of humans is motivated because tasks are harmful traditionally involve industrial tasks (heavy lifting, dangerous substances, and so forth). These replacements can be motivated because they minimize exposing humans to harm or danger. Although borderline cases may raise labor concerns, that is not something I discuss in this article.

However, more complicated cases involve, for example, the positive potential for replacing human sex workers with robots. It remains an empirically open question as to what extent replacement is possible. However, in cases where it is possible and in part in virtue of that justification, sex robots should be closed and limited information systems, rather than the cloud-based solutions that currently seem to exist on the market. Moreover, for any such example the functionality should be as limited as possible, to avoid some of the safety and security concerns raised in Sect. 2. I will return to examples of how this balance can be achieved in Sects. 3.3 and 4.

3.2 When AI Systems Are Better Than Humans

Sometimes AI systems are better than humans at performing a specific task. In such cases, if the task is sufficiently important, it may justify replacing humans with AI systems. For example, in the last five years, it has been found that the best result for

[43] An anonymous reviewer points out that many sex workers are satisfied with their jobs, not facing abuse, and would like to continue. This certainly complicates the replacement discussion in this case. However, I don't think this affects my overall argument. Indeed, what I suggest above is that sex robots might be one of the exceptions against replacing humans with social robots. Under the reviewer's argument, this would limit such replacements.

[44] See, e.g., Danaher and Mcarthur (2017).

[45] For obvious reasons, I will not link to any of the sites.

[46] Cf., e.g., Solove (2008).

the diagnosis of skin cancer is achieved through human-AI collaboration.[47] Nevertheless, it is not far-fetched to believe that, in the future, AI alone might surpass the combination, for some applications. Long-term, this may imply that we should replace humans with AI (i.e., even if, in the near future, we should expect a benefit from having a human involved). Moroever, if the combination of AI and humans already surpasses a human on their own, that already has implications for replacement.

However, the types of tasks for which AI systems seem to outperform humans are mostly non-social tasks, so these examples seem to provide no reason to favor social AI.[48] Although this may change over time, it is not clear if outperformance on purely social tasks is ever possible considering the arguments from Sect. 2.1; that is, we may value real social interactions irrespective of behavioral performance.[49]

Nevertheless, one potential reason for replacing humans with social AI systems would be to minimize the potential for bad actors.[50] However, given the arguments I have raised in Sect. 2.3, it seems that we have strong reason to question how this would resolve any of these problems. Keep in mind here that it may be beneficial to replace humans with AI and other machines for non-social tasks (e.g., to ensure a verifiable and audible process), but it is unclear how social AI systems as such provide an overarching benefit. Indeed, generative AI seems to provide a more efficient way for fraudsters to provide fraudulent material at low cost. As argued previously, if good social AI systems were common, this might result in a higher level of trust in social AI systems in general, and thereby increase the risk of successful fraud. Keep in mind how social AI systems in general, and LLMs in particular, may change the dynamics of email scams. Traditional advanced-fee scams are often designed in such a way that most people wouldn't fall for them. The reason for this is that once a human responds to them, the continued conversation is costly for the scammer as it involves human labor. With social AI systems, this cost could potentially be lowered radically and we could see an increase in more advanced scams.[51]

Lastly, we need to recognize that AI systems can currently outperform humans in the analysis of big datasets (e.g., auditing). However, while such examples do not require the systems to be social, they do have access to and control of large data sets, which poses informational worries. In these cases, we should, as previously noted, implement closed AI systems with strong data protection practices (e.g., encryption) and we should limit their ability to spread and disseminate information and data. I will return to examples of how this balance can be achieved in Sect. 4.3 and 5.

3.3 When AI Systems Are More Efficient Than Humans

Beyond situations when AI systems perform better than humans (i.e., when the systems outperform the best human, an average human, or a set of humans), AI systems

[47] See, e.g., Hekler et al. (2019); and Tschandl et al. (2020).

[48] There are some special exceptions to this. For example, pedagogical context involving children with autism (Tartaro & Cassell, 2008; Kim et al., 2013). This is in line with the exceptions I have in mind.

[49] Cf. Lundgren (manuscript).

[50] As mentioned in fn. 11, p. 5.

[51] See, e.g., Herley (2012) for a detailed analysis of the advanced-fee scams.

can also be more cost-efficient than humans—and if the system performance is sufficiently good, cost-efficiency can be a reason to replace humans with AI systems. This immediately raises the question of what cost-efficiency implies in the given context. For example, is cost-efficiency about the social function the system provides, or is it about something else? Here I understand the reason to replace humans with AI systems based on cost-efficiency in a very normal sense: that is, it is cost-efficient relative to the task we seek to accomplish (e.g., over its "lifetime", the AI system may prove less costly than a human, while producing the same outcome).

AI systems can increase cost-efficiency in the economy by replacing non-social jobs or functions. That is generally less problematic relative to the concerns that I have discussed in Sect. 2. However, AI systems may also prove cost-effective for jobs that have some social features, such as cashiers or in care work. Again, I propose that we should limit information usage, collection, and functionality, but this raises the question of whether such systems are possible. Although I recognize the trade-off between information usage and functionality,[52] I think there is a plausible path for reducing functionality and thereby reducing information usage.

Think of a cashier. While a human cashier may be very social—for example, in a small store in a small area with many regular customers—the work task does not require social engagement. It is possible to satisfy the tasks of a cashier with minimal information exchange. Clearly, in such cases, something may be lost, but I wonder how that is different from stores in which customers use a self-checkout instead of a human cashier. That is, the trade-off involved is not between humans and social AI systems, but it involves a tripartite choice between human, social AI systems, and AI systems with very limited information usage and processing. I think it should be clear that the social functionality of the systems does not override the concerns in Sect. 2 because the benefit provided is so limited. Rather we must consider whether we are re-designing a society with less social engagement and what the cost of that is (i.e., we should either go for the task-limited AI system or keep humans in place).

Another example is that of PARO. PARO is an AI-enhanced stuffed seal, which makes purring sounds and moves like a cat that enjoys being petted. PARO is designed to react to human touch and adapt to the way a given human touches it to create positive responses whenever an individual human is caressing it (i.e., unlike, for example, a cat, you cannot fail to pet PARO satisfyingly, since PARO will adapt to your petting-behavior). PARO has proven helpful in dementia care, for example, as a tool in therapy.[53] Despite some negative aspects of using PARO (such as the cost and resistance amongst some patient groups, particularly men), PARO is an excellent example of a limited information system. It is the type of device that can function without an Internet connection and with very limited informational inputs. Of course, PARO is arguably not an example of a robot that replaces humans, but rather a tool that can be used in conjunction with humans. Nevertheless, these examples illustrate the possibility of creating systems that do not override the concerns raised in Sect. 2 by being functionally limited information systems. The question at hand, then, is whether there are cases when we want to retain some social features and if that goal

[52] See, e.g., Lundgren (2021b).

[53] For reviews see, e.g., Hung et al. (2019); Kang et al. (2020).

can override the given concerns. However, given that this question overlaps with questions I will raise in the next section, I will address all these issues in tandem.

Lastly, so far, I have discussed systems as a whole, so it may be worthwhile to explain how we ought to handle cases in which a social AI system is more efficient than a human for singular tasks. For example, suppose that I want to share a song with a friend: doing so through Google Assistant or Siri would arguably be more efficient than doing it myself.[54] However, we cannot view these choices in isolation; we have to view the total cost of the systems and it is questionable whether these systems' efficiency outranks humans to the degree that their complete use overrides previous concerns. Moreover, the singular function is actually not that complicated for humans to perform. Indeed, major music tools standardly have simple sharing functions for songs and playlists.

4 Can and Should Social AI Systems Fill a Gap?

So far, I've discussed the issue of whether social AI systems should replace humans. But there is another issue: in the absence of humans, can—and if so, should—social AI systems fill a gap? That is, there may be tasks that a human currently does not currently—or cannot—undertake but that a social AI system could perform. Here we can think of "tools" such as PARO and the opportunity they provide in dementia care. However, we can also ask whether lonely people can be helped by robotic or other AI-based friends or even romantic partners. There might also be a place for AI systems in roles that are currently occupied by humans without replacing them (e.g., a robotic nurse that is available twenty-four-seven and can help whenever a human nurse is unavailable).

For some of these systems, we can make the argument that we can limit their social functionality without losing much of the desired functionality. That is, by using limited information systems that have minimal social interaction we can reap the benefits while avoiding the worries raised in Sect. 2. However, some of these applications would not function without being social (i.e., passing a certain threshold of informational exchange and producing various contextually appropriate behaviors). An example is AI-based friends. One may think that despite the potential downsides, an AI-based friend may be better than no friend for an individual that has no friends. However, here it is important to remember that while some of the worries I raised in Sect. 2 concern the individual end-user (e.g., it is about their privacy, their inauthentic social relations, and the risk that they may be manipulated, abused, or harmed), some worries concern us all (i.e., we know that information collection can have bad effects for all of us, e.g., through influences of democratic election).

Thus, in one sense it becomes a question of how to balance an individual's needs against the individual costs of using that system, and if that calculus is positive, we would have to weigh that against the whole collective. Many would think that we should prioritize those who are worst off (which friendless individuals arguably

[54] Although this is not technically a form of replacement in the sense that I am concerned with, it is an illustrative example reformulated from a comment by one of the reviewers.

would be a good example of). However, it is also important to keep in mind that we would be supplying a solution that is potentially very costly (given the concerns I raised in Sect. 2) for someone who is already socially vulnerable. This makes me wonder if this is the best we can do.[55]

As I have argued, we can avoid most of the challenges in Sect. 2 by focusing on using AI systems with very limited social functions (limited information collection, processing, and sharing).[56] Arguably such systems are also easier to develop, and they should be cheaper to produce and more cost-effective to use. If so, then we can expect that a discussion of using social AI systems is relevant only in a context in which we already have the potential to free up human labor in the economy (of course, this is a contingent truism, as that process has already started).

Thus, we can ask ourselves if the profits from human-AI replacement in other contexts mean that there may be purely social tasks that humans can fulfill. Of course, no one can be a romantic partner for a particular individual, in a genuine and ethically appropriate sense without a proper desire to be in a romantic relationship with that individual. However, in the broader scheme of things, as the presence of AI systems continues to increase in the labor market, we can ask whether we can expand human social tasks. Moreover, we need to ask ourselves if we can build a better society that is more fruitful for social interaction, and if that would create a conduit for social relations, rather than using AI systems to solve symptoms of our individual and collective failures. Nevertheless, one might think that we ought to provide both approaches. I do not think that would be successful and I provide three reasons (beyond the concerns raised in Sect. 2) against using social AI for lonely people:

First, in using social AI systems to provide company for lonely people, the individual and the society are only addressing the symptoms rather than the root causes of a social problem. That is, social AI systems provide a replacement for the lack of social interaction for a given individual, but they do not address or resolve the fact that there is a lack of social interaction for the given individual. This goes to the question of whether we can do better. As noted above, one might interject and argue that we can do both. However, although I don't doubt that there may be cases of loneliness that are best resolved—at least temporarily—by a social AI system (e.g., because the root cause is a chronic problem), I would worry that doing both is contraproductive. Indeed, these kinds of actions are not cost-free (in a manner of societal and individual resources), which means that these options will necessarily trade off each other. Of course, one might think that we ought to invest resources in temporary solutions while we wait for more permanent solutions and that the solutions therefore should be combined. Moreover, ignoring the fact that investments in temporary solutions drain resources that could have been used for solving the root causes, there are further problems with the dual approach: If the private sector is making money out

[55] As loneliness is a growing problem (see, e.g., Hosozawa et al. 2024), we should ask what is wrong in societies with increasing loneliness. Especially given the well-known medical implications.

[56] In some cases, certain memory functionality will be required. For example, a carebot used in dementia care might need memory functionality to ensure safety and good care. It might also need to be connected so that it can call for help in case of emergencies. That does not mean that we cannot apply limits on data usage and make use of suitable deletion standards. However, this is in line with the exceptions of using social AI systems in (health)care.

of loneliness, that creates market incentives to maintain the status quo. Even if those market incentives aren't abused to influence policy, it might still have effects in the political arena if social AI were to be successful in reducing the effects of loneliness and thereby minimizing the political incentives to remedy the problem.

Second, setting the above-mentioned conflicts aside, the use of social AI may be detrimental to solving the root problem. For example, a person with social anxiety may grow accustomed to interacting with an artificial agent and thereby avoid addressing their challenges, even if they could be resolved and would lead to a higher level of well-being. While this is an empirical question that requires long-term studies, these types of worries are recognized by empirical researchers.[57] Hence, *pace* Fröding and Peterson, I worry that social AI systems do not provide social training, but rather provide an escape route for social interaction for some people.[58] Although there is empirical uncertainty, in the absence of sufficient empirical studies, we should proceed with caution and, given the potential risks, the burden of proof should be on the proponents of social AI systems.[59]

Third, and relatedly, there is also the problem of so-called "product market traps"—that is, "a phenomenon whereby consumers prefer the product not to exist, but cannot avoid using it."[60] Market traps have been studied in the case of social media, for which users' social media consumption seems to be grounded more on a fear of missing out (TikTok 40%; Instagram 76%) and addiction (TikTok 33%; Instagram 11%) than on entertainment (TikTok 31%; Instagram 20%), information gathering (TikTok 16%; Instagram 15%), and productivity/convenience (TikTok 6%; Instagram 8%). This can be compared with Google Maps (8% fear of missing out, 23% information gathering, and 69% productivity/convenience). We might worry about similar effects for social AI.

Thus, even setting aside the worries raised in Sect. 2, it does not seem as if social AI systems are something we ought to aim for. Of course, there are special cases in which such systems would be justified. However, I think the upshot is that social AI systems should be used in very limited circumstances. I propose limiting their usage in the same sense that we limit medical treatments and related medical care—that is, we only apply them in exceptional cases when it is justified. Of course, as I recognize in Sect. 2.1, there may be harmful tasks that social AI systems could take over (e.g., if robots had the potential to diminish prostitution and the sex trade—granted the obvious suffering involved—then such functionality would be highly desirable and a good ground for their justification). However, social AI systems do not seem to create benefits that sufficiently justify their *broad* usage. Moreover, as previously stated, there seem to be no obvious reasons not to limit information usage, processing, and spreading. Indeed, social AI systems should be disconnected, separated, units.

[57] See, e.g., Pentina et al. (2023).

[58] Fröding et al. (2021).

[59] Following Hansson (2013), we can presume an overridable right against risk exposure. In this case, the risk involves substantiative epistemic uncertainty, making things a bit more complex. However, the argument would be that the burden of proof lies on the risk-creator because they need to establish that the right is overridable or otherwise set aside.

[60] See Bursztyn et al. (2023, p. 38).

Lastly, we might worry that the distinction between replacement and gap-filling is unclear. For example, suppose time with a human caregiver is slightly reduced in favor of several hours or 24/7 interaction with a carebot. I think these questions might fall under the scope of medical exceptions. Of course, similar arguments might apply in other cases, such as the example with cashiers. However, as I have argued, those systems can often be severely limited (similar to self-checkouts today), which means that they wouldn't raise the type of concerns and worries that I have addressed in this paper (i.e., they would be acceptable in the same sense as self-checkouts).

5 Which Social Codes Are Appropriate for Social AI Systems?

While I have argued against the broad usage of social robots, I also recognize that there may be situations in which these concerns are overridden—that is, when some AI-based social interaction is justified. However, contrary to some previous writers, I will argue that the default standard ought to be that AI systems should behave *professionally* rather than in a *friendly* way (i.e., setting aside those extremely rare cases when an AI-based friend might be motivated). Behaving in a friendly way is very contextual. It requires knowledge about the individual, which increases the risk of abuse by malevolent agents, the risk of being manipulated, risks related to informational transfers, and creates inauthentic relationships.

Behaving professionally, on the other hand, is a relatively well-codified (or codifiable) behavior that can be used to limit informational usage, focus the AI system on a specific task, avoid creating norms that make humans more sensitive to AI-based manipulation, and limit the risks of inauthentic relationships.

Of course, the behaviors do overlap mildly but the differences are more substantial and—in this case—ethically relevant. To exemplify and explain the similarities and differences more clearly, consider how professional behavior often is friendly, even if professional friendliness is very different from the friendliness you get from a friend. Friends inquire about each other, beyond what is needed for a given task. Professional behavior limits this inquiry to what is needed for the given task. Friends are private and adapt their behavior to the individual. Professional behavior is often codified and impersonal. Moreover, even in those cases when it is personal it is rarely private (except for, for example, a therapist).

The beauty of a professional code of behavior is that it is rule-based and the rules can be adapted to the appropriate requirements. For example, one can easily change professional codes to increase or decrease social engagement. While a cashier might ask a customer at the checkout "have you found everything you were looking for?", we rarely see that question being asked by a manual machine-based checkout.

While settling the precise norms we should follow is difficult and beyond what can be done within the limits of one article, my point here is that we need not override the concerns raised in the second section. Indeed, although we ought to limit the existence of social AI systems, it is also clear that we can construct social systems that limit the concerns I have raised.

Nevertheless, there are special cases of contexts in which ideas of friendly AI may seem more reasonable. For example, one may argue that this is necessary in situa-

tions involving care. However, as I have suggested above, professional behavior can be friendly. Yet, there might be medical reasons for having a carebot behave more like a friend, in limited circumstances. Arguably, situations that may require a different code of behavior are situations in which the human-AI interacts with humans with reduced agency (e.g., in dementia care). However, even in these cases, another type of behavior is more appropriate: simply put, in care situations, a robot (or any other AI-based system) should behave as if it were a caregiver. Friendly behavior is not always appropriate in such situations and what we want is simply another form of professional behavior: that of a caregiver.

Moreover, it is important to note that all social interactions require limits on informational usage. Arguably, as far as possible, social AIs should be closed, self-contained systems, rather than open, interconnected systems.[61] Simply put, rather than the recent trend of cloud-based solutions ranging from smart TVs to smart fridges under the banner of the Internet-of-things, social AI systems should be offline, disconnected units, with strong information protections and data deletion procedures. This provides a further argument against social AI systems since the ability of the state or organizations to monitor and ensure that the usage of AI is appropriate will improve if the total amount of systems is minimized.

Lastly, I turn to a practical problem. Making social AI systems that behave like friends is simply more difficult than making systems that follow certain given social codes or goals. Consider, for example, that Fröding and Peterson argue "that AIs should be programmed to *behave* in a manner that *mimics* a sufficient number of aspects of proper friendship".[62] If we read them verbatim, then programming should imply coding system behavior. But their ideas are too vague to code. Alternatively, if we allow for machine learning algorithms to develop as socially friendly AI systems, then their idea standardly needs to be expressed in terms of a goal, that an AI system can optimize against. It seems as if their proposal is simply too underdefined to be practically applicable. Indeed, under which conditions is the goal of being a friendly AI satisfied?

Of course, this may be a problem for part of my proposal as well. But arguably, professional behavior is more easily codified than friendly behavior and so is caregiver behavior, even if less than ordinary professional behavior.

6 Concluding Comments

As I have argued we can challenge the idea of replacing humans with AI systems—in social contexts—because social AI raises various ethical concerns that individually supply *pro tanto* reasons against using such systems. I have raised three concerns related to the value of authentic social relations, risks related to the usage of data and information, and risks due to malevolent agents, abuse, and misuse.

We might think that such reasons can be overridden in the sense that social AI may be justified in some limited cases despite these concerns. However, as I have argued

[61] Cf. Glancy (2012).

[62] Fröding, et al., (2021), p. 213).

there may be cases in which social AI systems are justified, but in such cases, we should enact severe limits on informational and data gathering, and to the extent that it is possible, we should also limit functionality. Because of these aims, I also suggest that friendliness is the wrong aim for most social AI. Social AI systems, when they are justified, should be guided by professional behavioral codes, rather than being aimed to behave like a friend.

Lastly, it needs to be recognized that my arguments are empirically sensitive. Indeed, we may have to adjust them to future technological progress. Relatedly, one may wonder what the practical impact of my arguments ought to be. Simply put, I think these insights should be part of regulations. We already see a partial overlap with some of my conclusions in, for example, the EU's AI Act.[63]

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Declarations

Conflict of interest BL is a member of the ethics advisory board for the Estonian Centre of Excellence in AI.

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[63] For a discussion on how ethical evaluations relate to regulations see, e.g., Lundgren (2023).

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